

Technology #138

Implement additive manufacturing
crack-resistant material inserts at rail
joints to reduce hydrogen absorption and
prevent fatigue fracture

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Implement additive manufacturing crack-resistant material inserts at rail joints to reduce hydrogen absorption and prevent fatigue fracture

WHY DOES IT MATTER?

Documented recurring problem 'rail steel fatigue fracture' (broken rails/joints, FRA cause family T2): 3 of 25 FRA Form 54 accidents in the retrieved window are broken-rail/joint-caused, derailing 17 cars total. Causes observed: 'Broken Rail - Base [Desc.Changed-Refer to Current/Prev.Guides]'; 'Mismatched rail-head contour'. LIMITS STAMPED INLINE: carrier-reported regulatory reports; cause codes are classifications of the initial report, NOT proven root causes; reportable-above-threshold events only (Art. XXI.5 analog).

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in mechanical_structural. 10-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 10-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Compare fatigue life of traditional thermite weldments versus weldments with crack-resistant material inserts under simulated hydrogen-co.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: buckling / kink margin tests across duty cycle. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design; superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead; K for catheter constraint conditions is MODEL_DERIVED; MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.; MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.; MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.; Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m⁴), E (Pa), epsilon (-), E (Pa), I (m⁴), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: sigma (Pa), sigma (Pa), P_cr (N) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Hydrogen absorption in pearlite steel (R260) during welding and operation reduces fatigue life and fracture toughness, leading to faster crack propagation and critical fracture under dynamic loads

POTENTIAL VALUE DRIVER: Addresses: Documented recurring problem 'rail steel fatigue fracture' (broken rails/joints, FRA cause family T2): 3 of 25 FRA Form 54 accidents in the retrieved window are broken-rail/joint-caused, derailling 17 cars total. Causes observed: 'Broken Rail - Base [Desc.Changed-Refer to Current/Prev.Guides]'; 'Mismatched rail-head contour'. LIMITS STAMPED INLINE: carrier-reported regulatory reports; cause codes are classifications of the initial report, NOT proven root causes; reportable-above-threshold events only (Art. XXI.5 analog).

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: mechanical_structural

Disciplines: mechanical engineering, structural analysis, fatigue and fracture, materials science

Architecture: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: Implement additive manufacturing crack-resistant material inserts at rail joints to reduce hydrogen absorption and prevent fatigue fracture

Subsystems:

?: load path / structural backbone - ?

?: actuation or compliance element - ?

?: guidance / bearing interfaces - ?

?: fatigue-critical features register - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Implement additive manufacturing crack-resistant material inserts at rail joints to reduce hydrogen absorption and prevent fatigue fracture

Key Physics: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

MEC-002: Section bending stress. Expression: $\sigma = M * c / I$. Applicability to this invention: CONDITIONAL; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design. Engaged variables: NOT ESTABLISHED.

MEC-003: Hooke's law (linear elasticity). Expression: $\sigma = E * \epsilon$. Applicability to this invention: CONDITIONAL; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead. Engaged variables: NOT ESTABLISHED.

MEC-001: Euler buckling load. Expression: $P_{cr} = \pi^2 * E * I / (K * L)^2$. Applicability to this invention: APPLICABLE; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL_DERIVED. Engaged variables: P_{cr} .

Assumptions:

- stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design
- superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead
- K for catheter constraint conditions is MODEL_DERIVED
- MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.
- MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.
- MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.
- Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m^4), E (Pa), epsilon (-), ϵ (Pa), I (m^4), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: sigma (Pa), sigma (Pa), P_{cr} (N) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED,

UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'rail steel fatigue fracture in service', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'kink/buckling below required push ability', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'fatigue crack initiation at stress concentrator', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'wear debris generation at sliding interfaces', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'joint separation under overload', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'set-release of superelastic shape after set', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Detection or material design must prevent service fracture under measured S-N curves across temperature excursions;	Source: F&O
DI-002	Mechanism driver (proposed)	Hydrogen absorption in pearlite steel (R260) during welding and operation	Mode: IIR
DI-003	Expected effect target	Extended service life of rail joints, reduced frequency of broken rail accidents;	Mode: IIR
DI-004	Domain design input pattern: delivery path applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering proposals)	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: activation applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering proposals)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: duty cycle applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering proposals)	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance standards foundation	ASTM F2516 (train design) pathway UNKNOWN (NEUTRAL PROPOSED)	NEUTRAL_PROPOSED
DI-008	Custodied observation: Mechanical Properties and Dependency of Fatigue Resistance on Temperature	666f3f10ab6c36da801e05e81dff8a548ba3ad8a14a09f6b212e4b465a9db3e7d	Source: F&O
DI-009	Custodied observation: Advances in Resistant Welding of Fiber-Reinforced Composites	66ef3f10ab6c36da801e05e81dff8a548ba3ad8a14a09f6b212e4b465a9db3e7d	Source: F&O
DI-010	Custodied observation: Non-Destructive Testing and Evaluation of Advanced Structural Composite Materials	66ef3f10ab6c36da801e05e81dff8a548ba3ad8a14a09f6b212e4b465a9db3e7d	Source: F&O
DI-011	Custodied observation: Current research trends in design of high-speed rail track	66ef3f10ab6c36da801e05e81dff8a548ba3ad8a14a09f6b212e4b465a9db3e7d	Source: F&O
DI-012	Custodied observation: Evaluation of Reliability of Steel Thermally Treated	66ef3f10ab6c36da801e05e81dff8a548ba3ad8a14a09f6b212e4b465a9db3e7d	Source: F&O

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	load path / structural backbone realizing Implement additional manufacturing crack-resistant architectural details	CONCEPTUAL	desired architectural details
DO-002	actuation or compliance element realizing Implement additional manufacturing crack-resistant architectural details	CONCEPTUAL	desired architectural details
DO-003	guidance / bearing interfaces realizing Implement additional manufacturing crack-resistant architectural details	CONCEPTUAL	desired architectural details
DO-004	fatigue-critical features register realizing Implement additional manufacturing crack-resistant architectural details	CONCEPTUAL	desired architectural details
DO-005	load path / structural backbone interfaces with actuation or compliance element: signal/energy/functional (dimensions, tolerances)	CONCEPTUAL	signal/energy/functional (dimensions, tolerances)
DO-006	actuation or compliance element interfaces with guidance / bearing interfaces: signal/energy/functional (dimensions, tolerances)	CONCEPTUAL	signal/energy/functional (dimensions, tolerances)
DO-007	guidance / bearing interfaces interfaces with fatigue-critical features register: signal/energy/functional (dimensions, tolerances)	CONCEPTUAL	signal/energy/functional (dimensions, tolerances)
DO-008	Device boundaries: patient/fluid inlet and outlet termination/control/data/pressure required by system	CONCEPTUAL	control/data/pressure required by system
DO-009	Flow/force/signal path geometry realizing the mechanism	UNKNOWN	material design, crack-resistant, surface finish
DO-010	Operating range for member slenderness ratio within which the mechanism holds its state	UNKNOWN	desired behavior
DO-011	Operating range for yield/ultimate strength within which the mechanism holds its state	UNKNOWN	desired behavior
DO-012	Operating range for applied cyclic load amplitude within which the mechanism holds its state	UNKNOWN	desired behavior
DO-013	Operating range for required run-out cycles within which the mechanism holds its state	UNKNOWN	desired behavior
DO-014	Operating range for articulation radius within which the mechanism holds its state	UNKNOWN	desired behavior
DO-015	Operating range for friction coefficient at guides within which the mechanism holds its state	UNKNOWN	desired behavior
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism in its operating position	REDESIGN	sensing, properly selected, decision thresholds (sensed values)
DO-017	Information path: sensing element -> decision/monitor	CONCEPTUAL	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in
DO-018	Fixture enabling: buckling / kink margin tests across duty cycle	UNKNOWN	rig concept named, build specific instrumentation selection; acceptance instrumentation
DO-019	Fixture enabling: fatigue life (run-out) testing — rig control	UNKNOWN	named, build specific instrumentation selection; acceptance instrumentation
DO-020	Fixture enabling: friction and wear characterization — rig control	UNKNOWN	named, build specific instrumentation selection; acceptance instrumentation

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: buckling/kink margin test across duty cycle	UNKNOWN	Unknown concept name; fixture specification; instrument selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
member slenderness ratio	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
yield/ultimate strength	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
applied cyclic load amplitude	UNKNOWN (no sourced value)	N	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
required run-out cycles	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
articulation radius	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
friction coefficient at guides	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
rail steel fatigue fracture	Fails by rail steel fatigue fracture	design service life of 50 years (10 ⁷ cycles) (required design work) (no known fatigue conditions (and	NO TEST DATA (required design work) (no known fatigue conditions (and	
kink/buckling below required	Fails by kink/buckling below required	compressions below ultimate capacity and supports capacity (no known buckling load due to (Am	NO TEST DATA (required design work) (no known buckling load due to (Am	
fatigue crack initiation	Fails by fatigue crack initiation	compressions below ultimate capacity and supports capacity (no known buckling load due to (Am	NO TEST DATA (required design work) (no known buckling load due to (Am	
wear debris generation	Fails by wear debris generation	compressions below ultimate capacity and supports capacity (no known buckling load due to (Am	NO TEST DATA (required design work) (no known buckling load due to (Am	
joint separation under force	Fails by joint separation under force	compressions below ultimate capacity and supports capacity (no known buckling load due to (Am	NO TEST DATA (required design work) (no known buckling load due to (Am	
set-release of superelastic	Fails by set-release of superelastic	compressions below ultimate capacity and supports capacity (no known buckling load due to (Am	NO TEST DATA (required design work) (no known buckling load due to (Am	

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
rail steel fatigue fracture in service	Fails by rail steel fatigue fracture in service	designer to provide NOT ESTABLISHED	Uncertainty in proposed operating conditions (Art. XbV)
kink/buckling below required	Fails by kink/buckling below required	pushability test to demonstrate additional capacity	Unknown capacity below required (Art. XbV)
fatigue crack initiation at stress concentration	Fails by fatigue crack initiation at stress concentration	stress concentration reduction	Unknown fatigue crack initiation (Art. XbV)
wear debris generation at sliding interfaces	Fails by wear debris generation at sliding interfaces	material pairing and surface condition	Unknown wear debris generation (Art. XbV)
joint separation under overload	Fails by joint separation under overload	load on a larger sized area	Unknown mechanical capacity, joint failure (Art. XbV)
set-release of superelastic shape	Fails by set-release of superelastic shape	pressure point at storage	Unknown set-release of shape (Art. XbV)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Compare fatigue life of traditional thermoplastic with proposed	Compare fatigue life of traditional thermoplastic with proposed	Compare fatigue life of traditional thermoplastic with proposed	Compare fatigue life of traditional thermoplastic with proposed
VF-002	run killer experiment: falsification test	run killer experiment: falsification test	run killer experiment: falsification test	run killer experiment: falsification test
VF-003	buckling / kink margin tests across duty cycle	buckling / kink margin tests across duty cycle	buckling / kink margin tests across duty cycle	buckling / kink margin tests across duty cycle
VF-004	fatigue life (run-out) testing. Acceptance	fatigue life (run-out) testing. Acceptance	fatigue life (run-out) testing. Acceptance	fatigue life (run-out) testing. Acceptance
VF-005	buckling/kink margin test across the duty cycle	buckling/kink margin test across the duty cycle	buckling/kink margin test across the duty cycle	buckling/kink margin test across the duty cycle
VF-006	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading
VF-007	buckling / kink margin tests across duty cycle	buckling / kink margin tests across duty cycle	buckling / kink margin tests across duty cycle	buckling / kink margin tests across duty cycle
VF-008	overload margin demonstration to worst case	overload margin demonstration to worst case	overload margin demonstration to worst case	overload margin demonstration to worst case

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	acceptance criteria to be fixed at a later point	Validation definition; validation is

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	nitinol / polymer composite membrane	minimally invasive device practice	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
nitinol / polymer composite material	nitinol / polymer composite material	1	Material	nitinol / polymer composite material	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- laser-cut hypotube / braided composite shaft: domain registry manufacturing route - candidate only; no process qualification exists
- passivation and surface finish control: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter member slenderness ratio [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter yield/ultimate strength [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter applied cyclic load amplitude [P_cr N]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter required run-out cycles [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter articulation radius [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter friction coefficient at guides [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}

- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}
- {'unknown': 'acceptance criterion for verification VF-008', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-008'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed test article	maintaining the proposed test article	maintaining the proposed test article	maintaining the proposed test article	maintaining the proposed test article
WP-02	constraint-compliance envelope article	per method	constraint-compliance envelope article	constraint-compliance envelope article	constraint-compliance envelope article	constraint-compliance envelope article
WP-03	article for VF-001	per method	Compare fatigue life of test article	Compare fatigue life of test article	Compare fatigue life of test article	Compare fatigue life of test article
WP-04	article for VF-002	per method	run killer experiment: test article	run killer experiment: test article	run killer experiment: test article	run killer experiment: test article
WP-05	article for VF-003	per method	buckling / kink margin test article	buckling / kink margin test article	buckling / kink margin test article	buckling / kink margin test article
WP-06	article for VF-004	per method	fatigue life (run-out) test article	fatigue life (run-out) test article	fatigue life (run-out) test article	fatigue life (run-out) test article
WP-07	article for VF-005	per method	buckling/kink margin test article	buckling/kink margin test article	buckling/kink margin test article	buckling/kink margin test article
WP-08	article for VF-006	per method	fatigue run-out at worst stress test article	fatigue run-out at worst stress test article	fatigue run-out at worst stress test article	fatigue run-out at worst stress test article
WP-09	article for VF-007	per method	buckling / kink margin test article	buckling / kink margin test article	buckling / kink margin test article	buckling / kink margin test article
WP-10	article for VF-008	per method	overload margin demonstration test article	overload margin demonstration test article	overload margin demonstration test article	overload margin demonstration test article

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 10-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.